

Odai aqlan

- Create internal and external customer tables and load data into them.
- Notice the issue related to delimiter inside the address column value, find a solution to make hive read the file properly without changing the separator. (Do some web search)

```
•create external table customer_external (  
  customer_id INT,  
  name STRING,  
  email STRING,  
  phone_number STRING,  
  address STRING,  
  join_date DATE,  
  start_date DATE,  
  end_date DATE,  
  is_current STRING  
)  
Row format serde 'org.apache.hadoop.hive.serde2.OpenCSVSerde'  
with serdeproperties (  
  "separatorChar" = ",",  
  "quoteChar" = "\""   
)  
stored as textfile  
location '/ext/cus/'  
tblproperties('skip.header.line.count'='1');
```

```
•create table customer_internal (  
  customer_id INT,  
  name STRING,  
  email STRING,  
  phone_number STRING,  
  address STRING,  
  join_date DATE,  
  start_date DATE,  
  end_date DATE,  
  is_current STRING  
)  
Row format serde 'org.apache.hadoop.hive.serde2.OpenCSVSerde'  
with serdeproperties (  
  "separatorChar" = ",",  
  "quoteChar" = "\""   
)  
stored as textfile  
tblproperties('skip.header.line.count'='1');
```

```
LOAD DATA LOCAL INPATH '/data/odai/customer_scd2_mixed.csv' INTO TABLE test.customer_external;
```

```
LOAD DATA LOCAL INPATH '/data/odai/customer_scd2_mixed.csv' INTO TABLE test.customer_internal;
```

I created two tables customer internal and external with all the customer columns.

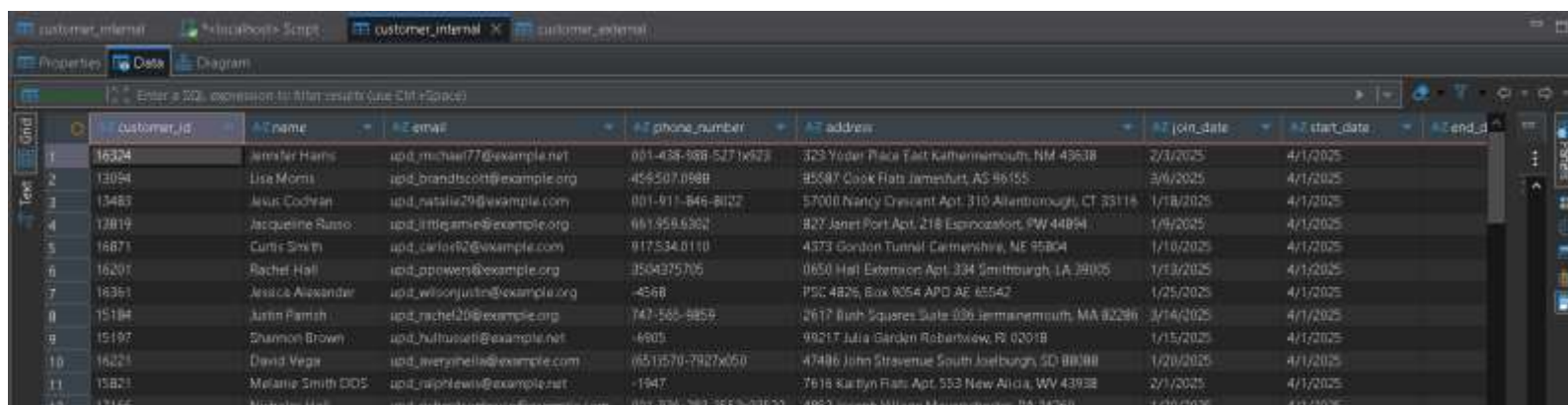
And because the address column sometimes has commas inside the text, I do some search and I ask Ai to have solution for this issue so I found **OpenCSVSerde** it's a tool in hive use in this situation ,so I use it with :

"separatorChar" = "," so Hive knows the fields are split by commas.

"quoteChar" = "\"" so Hive treats anything inside quotes as one value, even if it has commas.

I set the table to be stored as a text file and added skip.header.line.count=1 so Hive ignores the first line of the CSV (the header).

then, I loaded the data from my local file customer_scd2_mixed.csv into the two table



	customer_id	name	email	phone_number	address	join_date	start_date	end_date
1	16324	Wendy Harris	upd_michael77@example.net	001-438-908-5271x923	323 Yoder Place East Katonahamouth, NM 43638	2/3/2025	4/1/2025	
2	13094	Lisa Morris	upd_brandscott@example.org	456507.0988	85587 Cook Flats Jansenfurt, AS 86155	3/6/2025	4/1/2025	
3	13483	Jenna Cochran	upd_natalie29@example.com	001-911-846-8022	57000 Nancy Crescent Apt. 310 Allentownough, CT 33118	1/18/2025	4/1/2025	
4	13819	Jacqueline Russo	upd_jittigamie@example.org	661959.6362	827 Janet Port Apt. 218 Espronozafort, PW 44894	1/9/2025	4/1/2025	
5	16871	Curtis Smith	upd_carlos62@example.com	917534.0110	4873 Gordon Tunnel Carmenshire, NE 95804	1/10/2025	4/1/2025	
6	16201	Rachel Hall	upd_gpowers@example.org	3504375705	0850 Hall Extension Apt. 334 Smithburgh, LA 39005	1/13/2025	4/1/2025	
7	16351	Jessica Alexander	upd_wisoryjustin@example.org	4568	PSC 4826, Box 8054 APO AE 65542	1/25/2025	4/1/2025	
8	15184	Austin Parrish	upd_rachel20@example.org	740-565-6859	2617 Bush Squares Suite 306 Jernanemamouth, MA 82286	3/14/2025	4/1/2025	
9	15197	Shannon Brown	upd_hultuswift@example.net	4605	99217 Julia Garden Robertview, RI 02018	1/15/2025	4/1/2025	
10	16221	David Vega	upd_averyrhell@example.com	6512570-7927x050	47486 John Stravenue South Joelburgs, SD 88088	1/20/2025	4/1/2025	
11	15821	Melanie Smith DDS	upd_raphelex@example.net	1947	7618 Kaitlyn Flats Apt. 553 New Alicia, WV 43938	2/1/2025	4/1/2025	
12	17166	Nicholas Hall	upd_rubertobrooke@example.com	001-636-281-3553x21523	4852 Joseph Village Meyerschester, PA 34260	1/20/2025	4/1/2025	

Odai aqlan

- Try to drop both external and internal tables and notice what happens to the data.

```
DROP TABLE customer_external;  
DROP TABLE customer_internal;
```

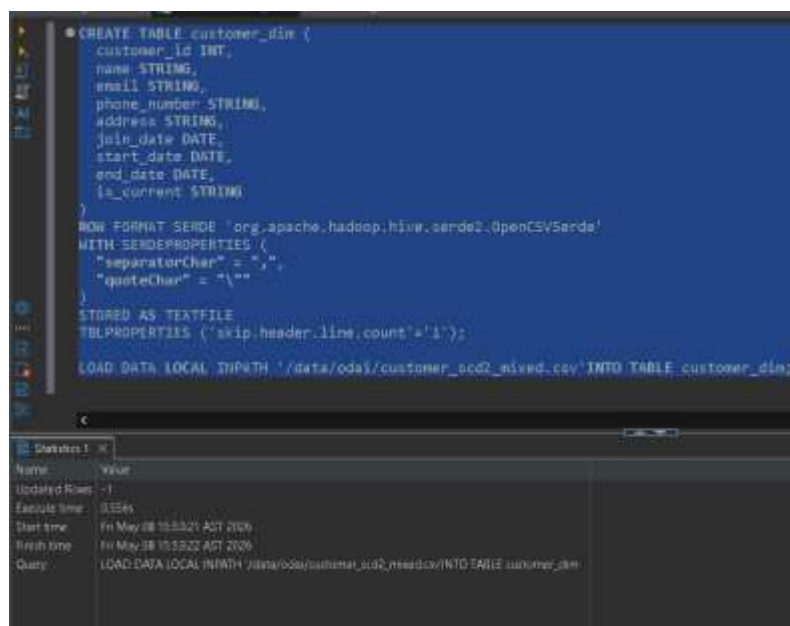
When I dropped the two table I saw the different,

In the internal hive deleted both the table definition and the actual data files

But When I dropped the external hive only removed the table definition (the metadata), but the data files in HDFS stayed in place

Because of that this is why external tables are useful when you want Hive to manage metadata but keep the raw data .

-
- Create customer dimension (Slowly changing dimension type 2) as “customer_scd2_mixed.csv” file.



```
CREATE TABLE customer_dim (  
  customer_id INT,  
  name STRING,  
  email STRING,  
  phone_number STRING,  
  address STRING,  
  join_date DATE,  
  start_date DATE,  
  end_date DATE,  
  is_current STRING  
)  
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'  
WITH SERDEPROPERTIES (  
  "separatorChar" = ",",  
  "quoteChar" = "\""   
)  
STORED AS TEXTFILE  
TBLPROPERTIES ('skip.header.line.count'='1');  
  
LOAD DATA LOCAL INPATH '/data/odai/customer_scd2_mixed.csv' INTO TABLE customer_dim;
```

Execution Results:

Item	Value
Name	Wale
Updated Rows	-1
Execute time	0.00s
Start time	Fri May 08 10:55:21 APT 2020
Finish time	Fri May 08 10:55:22 APT 2020
Query	LOAD DATA LOCAL INPATH '/data/odai/customer_scd2_mixed.csv' INTO TABLE customer_dim;

Here I created the customer dimension and load the data into the table

Odai aqlan

- Use “customer_updated.csv” file to insert new data with end_date = null and is_current = ‘1’ and update changed data to be with end_date and is_current = “0” to track history.
- “Hive doesn’t support update or deletes so find a work around to make this SCD type 2 without using Transactional tables”

I searched about the common approaches used for implementing Slowly Changing Dimension Type 2 in hive, most solutions rely on ETL workarounds.

From what I found, the **staging table approach** is the most widely used and practical method.

So the first I created the staging table and load customer updated data into the table

```
• CREATE TABLE customer_staging (  
  customer_id INT,  
  name STRING,  
  email STRING,  
  phone_number STRING,  
  address STRING,  
  join_date DATE  
)  
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'  
WITH SERDEPROPERTIES (  
  "separatorChar" = ",",  
  "quoteChar" = "\""   
)  
STORED AS TEXTFILE  
TBLPROPERTIES ('skip.header.line.count'='1');  
  
LOAD DATA LOCAL INPATH '/data/odai/customer_updated.csv' INTO TABLE customer_staging;
```

Statistics 1	Value
Time	
Updated Rows	-1
Execute time	0.114s
Start time	Fri May 08 15:59:07 AST 2026
Finish time	Fri May 08 15:59:07 AST 2026
Query	LOAD DATA LOCAL INPATH '/data/odai/customer_updated.csv' INTO TABLE customer_staging

Odai aqlan

Then now i use **insert overwrite** for the merge logic

```
● INSERT OVERWRITE TABLE customer_dim
SELECT
  d.customer_id,
  d.name,
  d.email,
  d.phone_number,
  d.address,
  d.join_date,
  CAST(d.start_date AS DATE) AS start_date,
  CASE
    WHEN s.customer_id IS NOT NULL THEN CAST(current_date() AS DATE)
    ELSE CAST(d.end_date AS DATE)
  END AS end_date,
  CASE
    WHEN s.customer_id IS NOT NULL THEN '0'
    ELSE d.is_current
  END AS is_current
FROM customer_dim d
LEFT JOIN customer_staging s
ON d.customer_id = s.customer_id

UNION ALL

● SELECT
  s.customer_id,
  s.name,
  s.email,
  s.phone_number,
  s.address,
  s.join_date,
  CAST(current_date() AS DATE) AS start_date,
  CAST(NULL AS DATE) AS end_date,
  '1' AS is_current
FROM customer_staging s;
```

This merge logic It simply overwrites the customer dimension table with a new version that combines old records and new updates.

Sample of customer dim data after merge:

	customer_id	name	address	start_date	end_date	is_current
1	12364	Larry Hess	556 Schmidt Gateway Apt. 111 Kaneton, NM 595		2026-05-08	0
2	12364	Larry Hess	556 Schmidt Gateway Apt. 111 Kaneton, NM 595	2026-05-08		1
3	12402	John Robinson	USNV Wagner FPO AA 72667		2026-05-08	0
4	12402	John Robinson	USNV Wagner FPO AA 72667	2026-05-08		1
5	12424	Trevor Brewer	56957 Jeanne Cove Suite 783 Krystalfort, PW 472		2026-05-08	0